Optimizing Online Retail Marketing Performance:

Behavioral Segmentation and Data-Driven Product Recommendation

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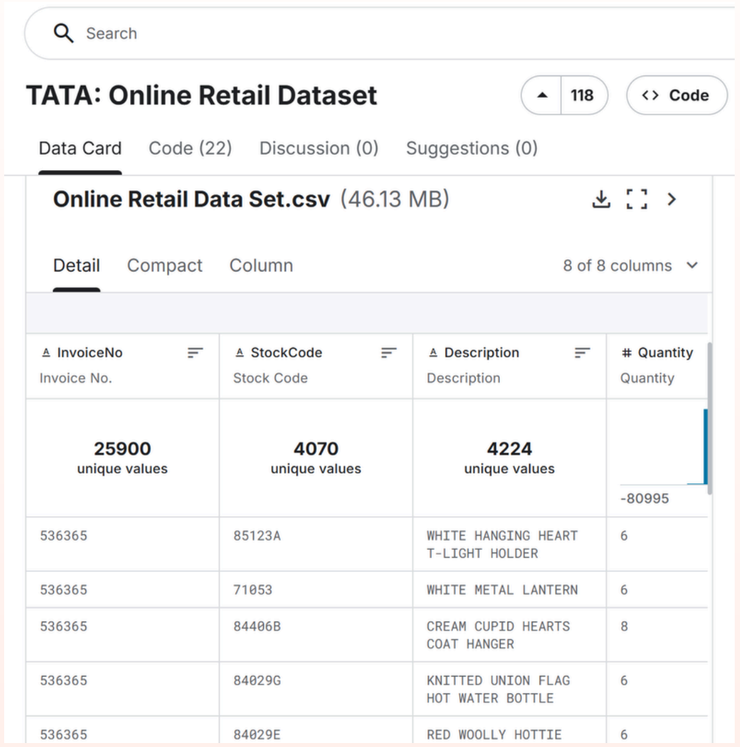
01

Background





▶▶ Background



Research Object

The subject of this study is a **UK-based non-store online retail business TATA** operating between 2010 and 2011, specializing in all-occasion gifts.
Despite generating significant transaction volume, the company has **no CRM strategy** and currently adopts a **product-centric "push" strategy** rather than a customer-centric approach, with a **"one-size-fits-all"** marketing model.

01

Inefficient Marketing Budget Allocation

- No mechanism to distinguish high-value VIP customers from one-time buyers
- Wastes resources on untargeted mass marketing instead of core customer retention

02

Missed Revenue Opportunities

- No systematic product recommendation system
- Fails to leverage transaction data for cross-selling & time-based repurchase prompts
- Limits Average Order Value (AOV) and Customer Lifetime Value (CLV)

02

Marketing Problems and Datasets





▶▶ This project aims to address two specific questions:



1

How can we segment the customer base to optimize marketing resource allocation?

2

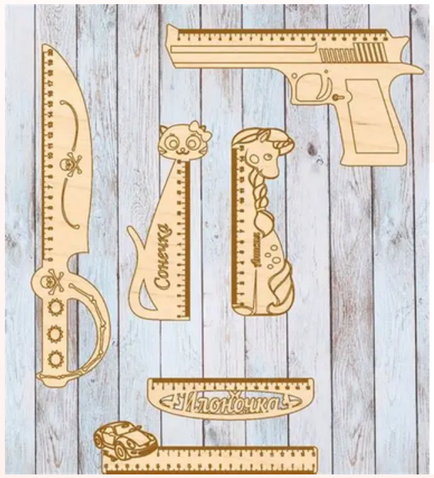
Which products should be recommended to customers to increase cross-selling and repurchase rates?

▶▶ Dataset Overview

	A	B	C	D	E	F	G	H
1	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
2	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010/12/1 8:26	2.55	17850	United Kingdom
3	536365	71053	WHITE METAL LANTERN	6	2010/12/1 8:26	3.39	17850	United Kingdom
4	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010/12/1 8:26	2.75	17850	United Kingdom
5	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010/12/1 8:26	3.39	17850	United Kingdom
6	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010/12/1 8:26	3.39	17850	United Kingdom
7	536365	22752	SET 7 BABUSHKA NESTING BOXES	2	2010/12/1 8:26	7.65	17850	United Kingdom
8	536365	21730	GLASS STAR FROSTED T-LIGHT HOLDER	6	2010/12/1 8:26	4.25	17850	United Kingdom
9	536366	22633	HAND WARMER UNION JACK	6	2010/12/1 8:28	1.85	17850	United Kingdom
10	536366	22632	HAND WARMER RED POLKA DOT	6	2010/12/1 8:28	1.85	17850	United Kingdom
11	536367	84879	ASSORTED COLOUR BIRD ORNAMENT	32	2010/12/1 8:34	1.69	13047	United Kingdom
12	536367	22745	POPPY'S PLAYHOUSE BEDROOM	6	2010/12/1 8:34	2.1	13047	United Kingdom
13	536367	22748	POPPY'S PLAYHOUSE KITCHEN	6	2010/12/1 8:34	2.1	13047	United Kingdom
14	536367	22749	FELTCRAFT PRINCESS CHARLOTTE DOLL	8	2010/12/1 8:34	3.75	13047	United Kingdom
15	536367	22310	IVORY KNITTED MUG COSY	6	2010/12/1 8:34	1.65	13047	United Kingdom
16	536367	84969	BOX OF 6 ASSORTED COLOUR TEASPOONS	6	2010/12/1 8:34	4.25	13047	United Kingdom
17	536367	22623	BOX OF VINTAGE JIGSAW BLOCKS	3	2010/12/1 8:34	4.95	13047	United Kingdom
18	536367	22622	BOX OF VINTAGE ALPHABET BLOCKS	2	2010/12/1 8:34	9.95	13047	United Kingdom
19	536367	21754	HOME BUILDING BLOCK WORD	3	2010/12/1 8:34	5.95	13047	United Kingdom
20	536367	21755	LOVE BUILDING BLOCK WORD	3	2010/12/1 8:34	5.95	13047	United Kingdom
21	536367	21777	RECIPE BOX WITH METAL HEART	4	2010/12/1 8:34	7.95	13047	United Kingdom









CustomerID: Unique identifier for the customer (4,338 identified customers).
InvoiceNo & InvoiceDate: Identifiers for the transaction session and time.
StockCode & Description: Identifiers for the specific product.
Quantity: Number of units purchased per transaction line. The median is 5 items, but ranges extensively.
UnitPrice: Price per unit in GBP. The median price is £1.95.
LineAmount (Derived): Quantity * UnitPrice, representing revenue per line item.
Country: Geolocation of the order, heavily dominated by the UK.

Key Features

- Strong seasonal peak from September to November
- 65.58% of customers are repeat buyers
- Highly skewed transaction: top 5% customers make >13 purchases

▶▶ Descriptive Statistics

Clean

指标	count
Quantity	396507.0
UnitPrice	396507.0
lineamount	396507.0

Full

指标	count
Quantity	397924.0
UnitPrice	397924.0
lineamount	397924.0

指标	count	mean	std	min	25%	50%	75%	max
Quantity	396507.0	20.82	290.52	-80995.0	2.0	5.0	12.0	80995.0
UnitPrice	396507.0	3.82	24.34	0.0	1.25	1.95	3.75	3897.0
lineamount	396507.0	15.96	161.53	-8715.5	3.75	9.95	23.35	16857.0

Input Variables	count	mean	std	min	25%	50%	75%	max
Recency	4335.0	93.21	100.17	1.0	18.0	51.0	143.0	374.0
Frequency	4335.0	4.25	7.64	1.0	1.0	2.0	5.0	206.0
Monetary	4335.0	2022.55	8923.22	0.0	305.77	668.11	1631.48	279138.02

We present descriptive statistics on **cleaned data** (excluding returns, postage, bank charges, and other non-product fees) to illustrate key distributional characteristics.

Transaction-Level Statistics

- Quantity, UnitPrice, and LineAmount show strong right-skewness: means are significantly higher than medians, with large standard deviations, driven by a small number of high-value transactions.
- Negative/zero minimum values confirm raw data contains returns and fee records, validating the necessity of data cleaning.

Customer-Level Statistics

- Frequency and Monetary are also right-skewed (means far exceed medians, large gaps between max and median values), indicating a small subset of customers have very high purchase frequency and spending, while most are low-frequency/low-spending.
- Recency distribution shows considerable dispersion.

03

Decomposition and Descriptive Model





▶▶ Descriptive Analysis Model & Framework

Descriptive Model

Input Variables (Customer History): Recency (days since last purchase), Frequency (number of purchases), Monetary Value (total spend).
I**ntermediate Variable (Segmentation):** Customer Tier (e.g., VIP, Potential Loyalist, Inactive).
Marketing Action (Intervention): Targeted Product Bundles and Personalized Email Recommendations.
Outcome Variables (Performance): Repurchase Rate, Average Order Value (AOV), and Customer Retention.

Analytics Methods Applied

Descriptive Analytics (Cluster/Segmentation Analysis): Used to translate raw transaction history into distinct customer segments (RFM model).
Prescriptive Analytics (Association Rule Mining): Used to identify complementary products (Apriori algorithm) to generate actionable recommendation lists.

04

Customer Segmentation via RFM Analysis





▸▸ Customer Segmentation via RFM Analysis

Purpose: to address the problem of <u>inefficient budget allocation.</u>
By understanding the heterogeneity of the customer base, we can identify high-value customers for retention and low-value customers for low-cost reactivation.

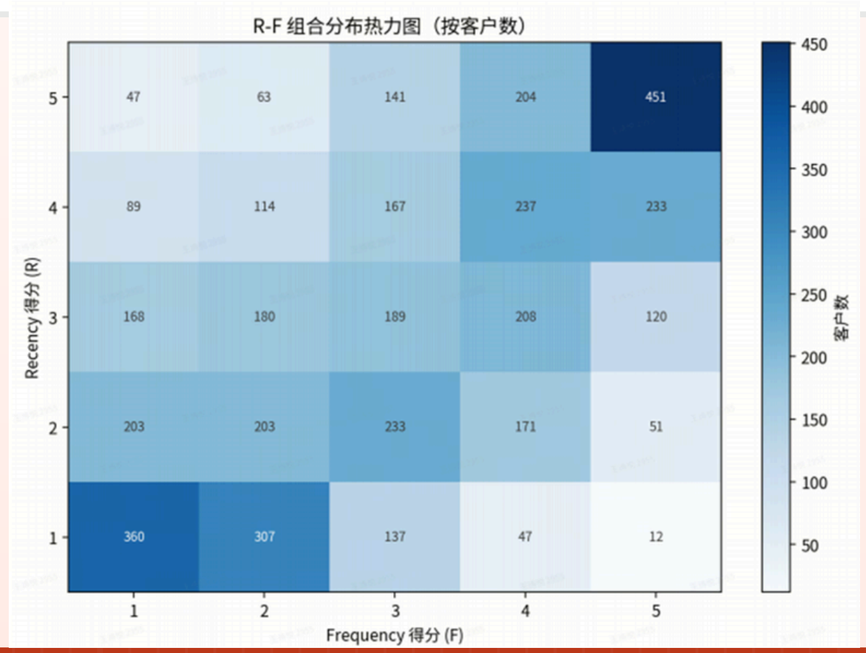
Data Cleaning
- Remove records with missing Customer ID/product description
- Filter out returns, postage fees, bank charges to retain real product sales

RFM Metric Calculation
- Compute Recency, Frequency, Monetary for each customer

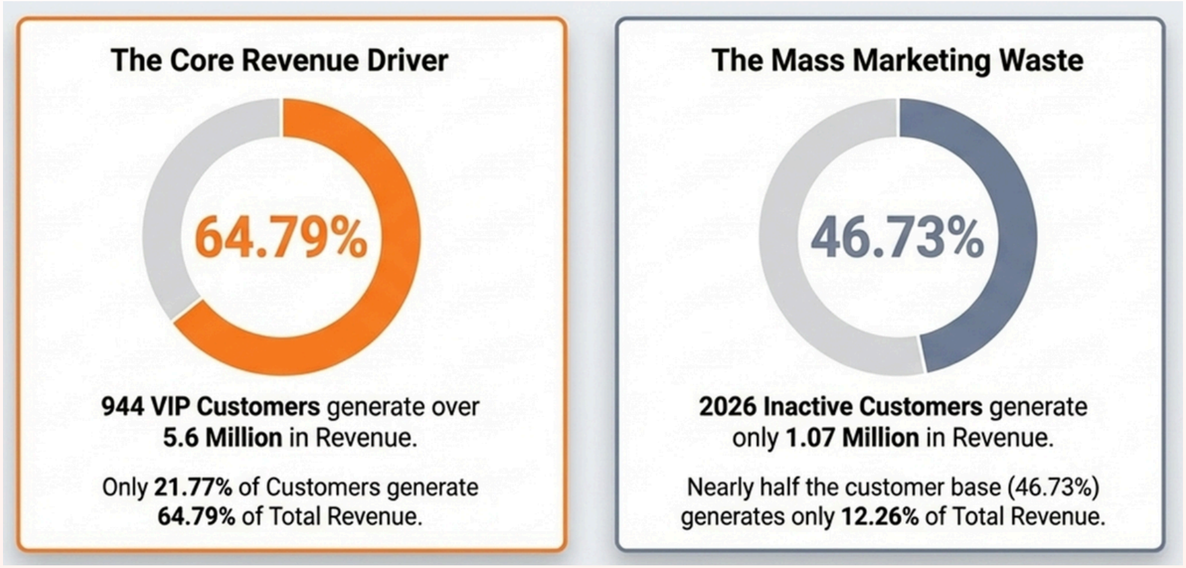
RFM Scoring (1–5 Scale)
- R：Reverse quintile scoring; F, M：Quintile scoring

Segmentation Rule
- Total score = R_score + F_score + M_score (3–15)
- Conditional mapping by quartiles to isolate real VIPs

Segment	Condition	Business Meaning
VIP	R ≥ 4 AND F ≥ 4 AND M ≥ 4	Recent, frequent, high-value core customers
Consistent Buyers	R ≥ 3 AND F ≥ 4 AND M ≥ 3	Recent, frequent loyal buyers
Potential Loyalists	R ≥ 2 AND F ≥ 3 AND M ≥ 2	Some purchase behavior, growth potential
Inactive	Otherwise	Low-frequency, low-spending, or chumed customers

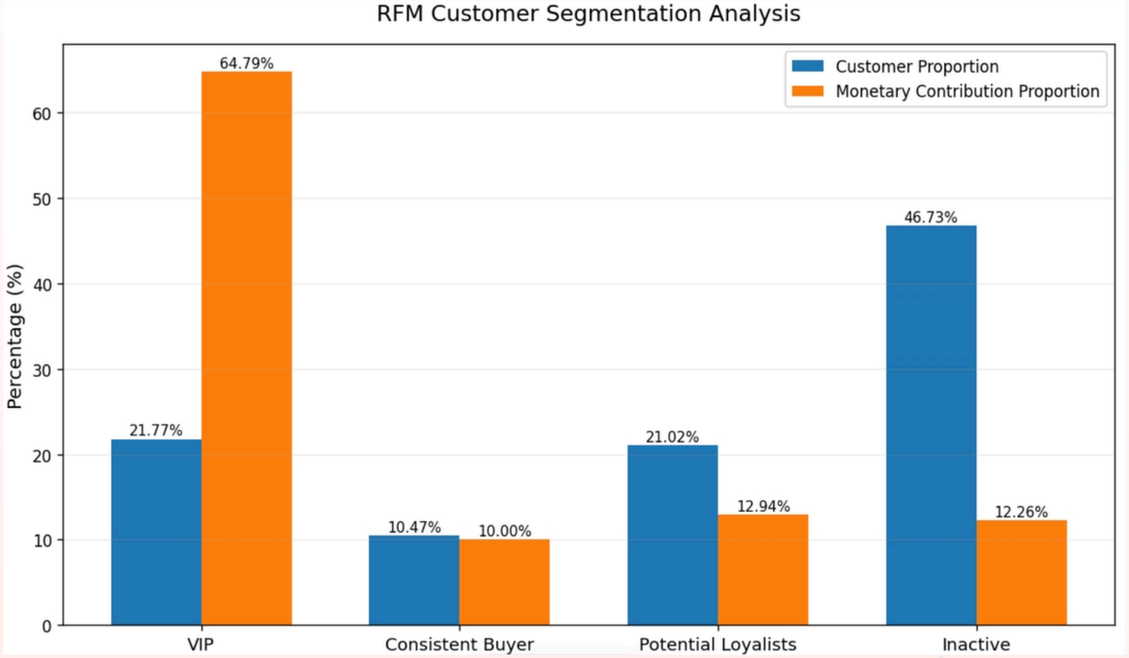


▸▸ Customer Segmentation via RFM Analysis



The results strongly validate the Pareto principle within the dataset. The “one-size-fits-all” strategy disproportionately under-serves the highly profitable VIP segment while wasting resources on the inactive majority.

Highly active core group driving the vast majority of revenue



High concentration of legacy one-time buyers (Inactive segment)

05

Product Recommendation via Market Basket Analysis





▶▶ Market Basket Analysis

Purpose: addresses the missed cross-selling opportunities by identifying hidden patterns in purchasing behavior to recommend the right products to the right customer segments.

Algorithm: Apriori association rule mining (Max Length = 5 items).

Stratified Mining Architecture

- **Tier 1:** Segment-specific rule mining (Threshold Tuning)
  - VIP & Consistent → Precision.
  - Inactive & Potential Loyalists→ Coverage.
- **Tier 2:** Global rule base as fallback (prevent recommendation gaps)

Validation Method: Leave-Last-Out (avoid data leakage, simulate real prediction scenario)

Segment Group	Min Support	Min Confidence	Business Logic
VIP & Consistent	0.015	0.18	Stricter Rules: Ensure high relevance for premium users. Avoid noise.
Inactive & Potential	0.005	0.1	Looser Rules: Capture weak patterns in sparse data to avoid "no recommendation".
Global (Fallback)	0.01	0.15	Safety net for cold-start scenarios.

▶▶ Market Basket Analysis






Recommendation Example:

Customer ID	Customer Status	Past Purchased Stock Codes	Recommended Stock Codes	Hit Stock Codes
12793	Inactive	21992, 22383, 20728	20725	20725
14670	Inactive	21922, 22967, 22804	20675, 20676, 22522, 22578	20675, 20676
14465	Inactive	22386, 85099B, 21931	22383, 22634, 82581, 84879, 85123A	22634, 85123A
17991	Inactive	22697, 22698	22423, 22699	22423
14304	Inactive	85049G, 22311, 21591	22138	22138

CustomerI	Segment	Last_Invoic	Last_Bought_Item	Recommended_Ite	Recommendation_Details
12347	VIP	581180	['23497', '23552', '	['20724']	20724 (置信度:0.53)
12349	Inactive	577609	['23112', '23460', '	['21094', '84879', '2	21094 (置信度:0.81); 8487
12354	Inactive	550911	['23201', '23204', '	['22634', '22383']	22634 (置信度:0.69); 2238
12357	Inactive	574740	['22064', '21232', '	['20675']	20675 (置信度:0.73)
12359	Consistent	C580165	['22797', '23245', '	['22722']	22722 (置信度:0.25)
12360	Potential L	571705	['21559', '84997A',	['84997D', '22629',	84997D (置信度:0.85); 226
12361	Inactive	545017	['22630', '22326', '	['20728']	20728 (置信度:0.62)
12362	VIP	580979	['23238', '23237', '	['23200', '23209', '2	23200 (置信度:0.43); 2320
12363	Inactive	563947	['23236', '21210', '	['22909']	22909 (置信度:0.75)
12370	Consistent	571905	['22784', '22465', '	['22570', '21136', '2	22570 (置信度:0.57); 2113
12371	Inactive	570521	['22915', '22924', '	['22748', '22746', '2	22748 (置信度:0.80); 2274
12374	Inactive	576345	['72741', '85174', '	['22835']	22835 (置信度:0.68)
12377	Inactive	542586	['21206', '22197', '	['22745', '22748', '4	22745 (置信度:0.83); 2274
12378	Inactive	562046	['22747', '85227', '	['22727', '22630', '2	22727 (置信度:0.79); 2263
12380	VIP	577180	['23480', '21731', '	['20725', '85099B', '	20725 (置信度:0.67); 8509
12390	Inactive	567530	['23126', '22326', '	['22748', '22556', '2	22748 (置信度:0.87); 2255
12391	Inactive	577228	['22669', '22668', '	['22383']	22383 (置信度:0.59)
12394	Inactive	569997	['23245', '23295', '	['23294']	23294 (置信度:0.83)
12397	Inactive	574336	['22374', '23203', '	['85123A']	85123A (置信度:0.69)
12398	Inactive	572567	['21125', '37495', '	['84997C', '22720', '	84997C (置信度:0.72); 227
12402	Inactive	541675	['20712', '22663', '	['22629']	22629 (置信度:0.67)
12405	Inactive	559928	['23241', '23240', '	['21212']	21212 (置信度:0.7

▶▶ Market Basket Analysis

Overall Performance:

- Hit Rate: 20.37% (Top-5 recommendations matched actual next purchase).
- Validation Sample: 1,296 effective customers.
- Coverage: 42.2% (1,830 out of 4,335 active users received valid recommendations).

Segment Breakdown (The "Rich get Richer" Effect):

- High Performers:
  - Consistent Buyers: 56.4% Coverage
  - VIP: 55.4% Coverage
- Challenge Areas:
  - Inactive: 40.0% Coverage
  - Potential Loyalists: 26.5% Coverage

20.37%	1,296	Top-5
Hit Rate	Effective validation sample size	recommended items

Marketing Recommendation Coverage Statistics

Metric	Result
Total Active Customers	4,335
Customers with Successful Recommendations	1,830
Overall Recommendation Coverage	42.2%
Number of Recommended Items per Customer	Top-3

Segment	Total Customers	Customers with Recommendatio	Coverage Rate
Consistent Buyers	454	256	56.4%
VIP	944	523	55.4%
Inactive	2,026	810	40.0%
Potential Loyalists	911	241	26.5%
Total	**4,335**	**1,830**	**42.2%**

06

Conclusions and Recommendations





▶▶ Conclusions and Recommendations

Key Findings

- **Inefficiency of "One-Size-Fits-All":** Analysis proves ~22% of customers generate ~65% of revenue. Budget should be disproportionately allocated to retaining this cohort via premium services, while low-cost, automated emails should target the 47% Inactive segment.
- **Predictive Accuracy:** Stratified Apriori association rules achieve a 20.37% hit rate for predicting next purchases. The system successfully generated actionable Top-3 product lists for 42.2% of the active user base.





Actionable Insights

- **For VIPs & Consistent Buyers:** Deploy the generated segment-specific recommendations immediately. Their high coverage (>55%) and strict rule thresholds ensure high relevance, driving incremental cross-sell revenue.
- **For Potential Loyalists:** Use association rules where available, but fill the gap with "Trending in Category" items to increase coverage and engagement.
- **For Inactive Customers:** Use the broader rules (lower thresholds) to identify "win-back" products, combining these with discount incentives could improve reactivation rates.

▶▶ Conclusions and Recommendations





Remaining Challenges

- **Data Sparsity:** "Potential Loyalists" segment shows only a 26.5% recommendation coverage rate.
- **Seasonal Trends:** Current analysis ignores strong holiday trends discovered during the EDA phase.

Solutions

- **Incorporate Seasonality:** Introduce time windows into the Apriori training set to create "Holiday Season" recommendation rules.
- **Advanced Modeling:** Explore Collaborative Filtering to supplement the Apriori rules.
- **Prescriptive Evaluation:** Use A/B testing framework to evaluate the true lift in AOV for recommendations.





Q&A Session

Group 6